Yana Lishkova

Pasadena, CA, USA | lishkova[at]caltech.edu | yanalish.github.io/yl-site-2026 | Google Scholar

Postdoctoral Scholar at Caltech developing optimization-based guidance and control for spacecraft and robots with formal safety guarantees, from theory to software- and hardware-in-the-loop (SIL/HIL) validation.

Education

University of Oxford, DPhil (PhD) in Engineering Science, Oxford Control Group — *2019 – 2024*

University of Cambridge, Aerospace and Control Engineering, Christ's College — *2015 – 2019*

- MEng in Instrumentation & Control and Aerospace & Aerothermal Engineering: *Honours with Distinction.*
- BA (MA Cantab) in Engineering: *Upper Second-Class (2:1).*

University of Edinburgh, Diploma of Higher Education in Science and Engineering — *2013 – 2015*

High School of Mathematics and Natural Sciences, Blagoevgrad, Bulgaria — *2008 – 2013*

Research and Industrial Experience

Postdoctoral Scholar Research Associate, Caltech, AMBER Lab (Prof. A. D. Ames) — *Sept 2025 – Present*

Postdoctoral Research Assistant, University of Oxford (EPSRC Doctoral Prize) — *Oct 2024 – May 2025*

Research Intern, Mitsubishi Electric Research Laboratories, Cambridge, MA — *Jun 2023 – Oct 2023*

DPhil Researcher, University of Oxford (Profs. Mark Cannon and Sina Ober-Blöbaum) — *Oct 2019 – Mar 2025*

MEng Researcher, University of Cambridge (Prof. M. Juniper) — *Oct 2018 – Jun 2019*

Summer Intern, Engineering Rotation, Marshall Aerospace and Defence Group — *Jul 2018 – Sep 2018*

Summer Intern, Mechanical Systems and Design, Sagentia — *Jun 2017 – Aug 2017*

Summer Intern, Development and Innovation, YASA Motors — *Jun 2016 – Sep 2016*

Research Themes and Publications

From theory to hardware: efficient methods for safe autonomy of spacecraft and robots, built on optimization and structure-preserving methods. First author on all 9 papers (1 journal, 8 conference); 2 papers under review; 1 journal paper ready for submission; 1 patent application; an invited ACC 2025 tutorial and 3 invited talks. Full list under Research on my website.

Safe spacecraft autonomy, guidance and control — *Caltech, Oxford*

- **Multi-agent space proximity operations:** lead an AFOSR-funded project on increasing the autonomy of multi-spacecraft close-proximity operations while guaranteeing safety.
- **Coupled 6DOF spacecraft control:** an exact difference-of-convex (DC) decomposition of the coupled position and attitude dynamics enables convex model predictive control (MPC) of both, validated on an ATMOS free-flyer.
  - **Y. Lishkova** et al., *Title to appear after review process*, submitted to IEEE ICRA 2027.
- **Flexible spacecraft and electrodynamic tethers:** optimal slewing while damping vibrations; a variational 6DOF model enables an electrodynamic-tether spacecraft transfer in the circular restricted three-body problem using attitude torques alone.
  - **Y. Lishkova** et al., *Multirate variational methods for simulation and optimal control of flexible spacecraft*, IFAC WC 2026. ORA
  - **Y. Lishkova** et al., *Variational approach for modelling and optimal control of electrodynamic tether motion*, IAC 2023. IAF
    - Invited article: **Y. Lishkova** et al., *Multirate discrete mechanics and optimal control for a flexible satellite model*, *GAMM Rundbriefe* 2024. GAMM
  - **Y. Lishkova** et al., *A multirate variational approach to simulation and optimal control for flexible spacecraft*, AAS 2020. arXiv

Safe exploration and landing with contingencies — *Caltech, MERL*

- **Divert-feasible lunar landing:** fuel-optimal descent with probabilistic guarantees of safe divert, using stochastic controllable sets computed offline via lossless convexification.
  - **Y. Lishkova** et al., *Divert-feasible lunar landing under navigational uncertainty*, IEEE CDC 2024. IEEE Xplore
    - **Y. Lishkova** et al., US Patent Application 18/975,196 (US 2026/0159256 A1), MERL. Google Patents
- **Contingency filters** steer toward a chosen target while keeping a required number of backup options reachable, without integer programming or combinatorial blow-up, using control barrier functions (CBFs), control Lyapunov functions (CLFs) and Hamilton–Jacobi reachability (HJR).
  - **Y. Lishkova** et al., *Steering with contingencies: combinatorial stabilization and reach-avoid filters*, IEEE CDC 2026. arXiv
- **Resilient multi-robot connectivity:** matrix CBFs keep a robot team's network connected and, after a robot fails, restore resilient connectivity within a set recovery time; tested on Crazyflie quadrotors.
  - P. Ong, **Y. Lishkova** et al., *Title to appear after review process*, submitted to IEEE ICRA 2027.

Robust and nonlinear model predictive control — *Oxford*

- **Successive convexification:** a DC model makes each MPC step a convex program, ensuring recursive feasibility and quadratic stability with or without disturbances, and convergence to a local optimum without them.
  - **Y. Lishkova**, M. Cannon, *A successive convexification approach for robust receding horizon control*, IEEE TAC 2025. IEEE Xplore

Y. Lishkova, M. Cannon, *Optimization-based trajectory generation and receding horizon control for systems with convex dynamics*, IEEE CDC 2023. IEEE Xplore

- **Multirate variational MPC:** tube-based nonlinear MPC discretising slow and fast dynamics at their own rates, shrinking the optimization problem for large computational savings at similar accuracy, with recursive feasibility.
  Y. Lishkova et al., *A multirate variational approach to nonlinear MPC*, ECC 2022. IEEE Xplore

Learning dynamical models — *Cambridge, Oxford*

- **Discrete Lagrangian neural networks:** learn a discrete Lagrangian and its symmetry group from observed motion, keeping predictions physically consistent and revealing conserved quantities.
  Y. Lishkova et al., *Discrete Lagrangian neural networks with automatic symmetry discovery*, IFAC WC 2023. IFAC-PapersOnLine
- **DC neural networks:** learned spacecraft dynamics in DC form for robust tube MPC (ongoing).

Structure-preserving numerical methods — *Oxford, FAU*

- **Multirate variational methods** for simulation and optimal control that preserve physical properties such as momentum and symplecticity, while efficiently handling dynamics on different timescales.
  Y. Lishkova et al., *Multirate discrete mechanics and optimal control: review and extension*, ready for submission.
- **Extensions and applications in the papers above:**
  - *Applied to flexible spacecraft:* slewing with vibration suppression (AAS 2020, GAMM 2024, IFAC 2026).
  - *Extended to constrained systems:* quaternion-constrained 6DOF electrodynamic-tether model (IAC 2023).
  - *Used to formulate multirate MPC:* smaller problems, computational savings, recursive feasibility (ECC 2022).
  - *Embedded in learning:* discrete Lagrangian networks learning models and their symmetries (IFAC 2023).

Testbeds: from algorithms to hardware — *Caltech*

- **ATMOS free-flyers:** open-source air-bearing robots that glide almost frictionlessly on compressed-air thrusters, recreating planar spacecraft motion; I co-lead our lab's ATMOS testbed for SIL/HIL validation. ATMOS
- **Crazyflie quadrotors:** palm-sized open-source drones flown as a team to test multi-robot algorithms. Bitcraze

Earlier aerospace projects — *Cambridge*

- **Aerofoil shape optimisation (MEng):** adjoint-based design, Fortran 90 flow solvers, automatic differentiation.
- **Martlet 3 rocket (Cambridge University Spaceflight, 2017):** two-stage boosted dart, Black Rock Desert.

Technical Skills

- **Control and numerical methods:** robust, tube and multirate MPC; lossless, successive and DC convexification; stochastic controllable sets; CBFs; HJ reachability; variational integrators and optimal control.
- **Robotics and hardware:** ROS 2, PX4, Gazebo, SIL/HIL testing, ATMOS free-flyers, Crazyflie quadrotors; testbed integration, PCB layout and debugging, CATIA, SolidWorks, COMSOL.
- **Software:** Python (JAX, CVXPY, NumPy, SciPy), MATLAB, Git, LaTeX; beginner in C++ and Julia.

Awards and Funding

- **SURF Mentor "Shout Out"** – Caltech, nominated by my summer research students — *2026*
- **AFOSR funding** – safe spacecraft autonomy, from a joint white paper with Prof. Ames — *2025 – present*
- **EPSRC Doctoral Prize** – personal postdoctoral fellowship at Oxford (£35k) — *2024*
- **RAeS Young Person's Achievement Commendation** – youth engagement in aerospace — *2023*
- **Amelia Earhart Fellowship** – Zonta International, US$10,000 — *2022*
- **EPSRC DTA Research Studentship** – full funding for my DPhil at Oxford — *2019*
- **M.R. Lynch Prize in Engineering** – University of Cambridge, final-year distinction — *2019*
- **Christ's College Graduate Scholarship** – Christ's College, Cambridge — *2019*

Selected Experience Beyond Research

- **Mentoring:** six AMBER Lab volunteers and interns (Caltech SURF and Connect); Master's thesis co-mentor.
- **Teaching:** Teaching Assistant and Lab Demonstrator, Oxford Engineering (2020 – 2024); OxNet access tutor.
- **Service:** reviewer for IEEE TCST, Automatica, Journal of Computational Dynamics, L4DC and CDC.
- **Leadership:** Junior Dean, St Hugh's College (2021 – 2025); Graduate Committee Chair (2022 – 2023).
- **Organiser:** lead co-organiser of GenSTEM (2023, 2024) and of an aerospace seminar series (2021 – 2023).
- **Outreach:** RAeS Oxford Young Person Representative; OxWoCS committee; Girls Who ML helper.

More about my students, teaching and outreach on my website.

Personal Details

- Dual Bulgarian and British citizen.
- Bulgarian and English (fluent); German (intermediate, TestDaF).
- Solo glider (sailplane) pilot since 2025; Oxford Half Blue.